🎯 UN SDG 2 (Zero Hunger) | 🌱 UN SDG 12 (Responsible Consumption) | ⚡ 15–30 Min Rescue Velocity

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A Hyper-Local, Crowd-Sourced Food Rescue & Safety-Compliant Redistribution Platform for Dhaka City

TARGET PILOT ECOSYSTEM

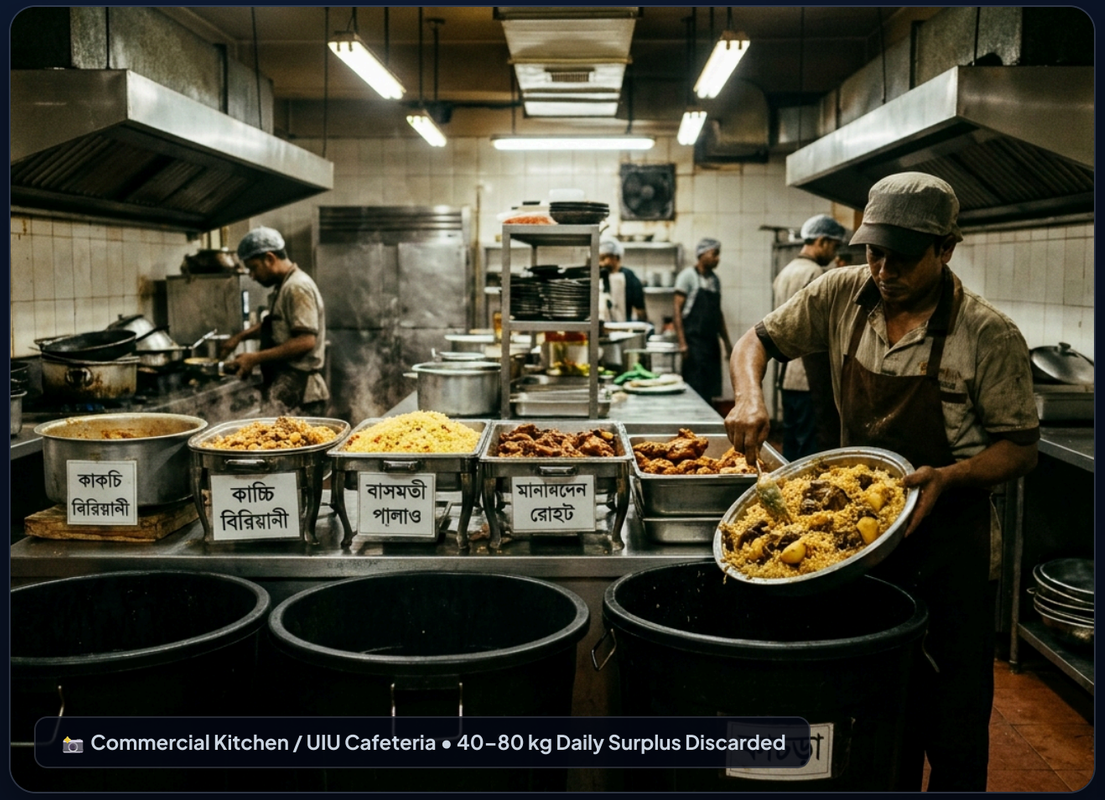
United International University (UIU) Central Cafeteria, Local Caterers & Satarkul/Badda Shelters

CORE SOFTWARE ENGINEERING GOAL

Eliminate cooked food dumping via sub-60s listing, proximity radar GIS, and zero-trust OTP handovers.

1. Problem Identification: What Problem Are We Solving?

Dhaka's critical paradox: Mountains of fresh cooked food discarded nightly while street children sleep hungry



📸 Commercial Kitchen / UIU Cafeteria • 40–80 kg Daily Surplus Discarded

🗑️ 1. The Daily Midnight Dump

University cafeterias (e.g. UIU), event halls, and restaurants cook in bulk. Every night, huge pots of hygienic, edible food are thrown straight into dumpsters simply because donors lack an instant charity contact.

⏱️ 2. The 4-Hour Biological Clock

Cooked food enters the bacterial danger zone (5°C–60°C). Under **BFSA safety guidelines**, food must be consumed within 4 hours. Manual phone calls and bureaucratic arrangements take 6+ hours—resulting in total spoilage.

🚫 3. The Missing Micro-Logistics Link

Local orphanages and shelters have zero transport vehicles or fuel budgets at 10 PM. While 100 children starve only 2 km away, there is no automated micro-logistics platform to dispatch student couriers to bridge the gap.

Root Cause: Communication failure & lack of automated hyper-local dispatch, NOT food shortage.

1. Evidence & Data: Backed by Hard Statistics

Quantifying the urban food crisis, biological decay limits, and social urgency in Dhaka

10.7M Tons
Annual Food Waste
UNEP Food Waste Index 2024 (BD)

4 Hours
Biological Safety Limit
BFSA Danger Zone (5°C – 60°C)

3.8 Million
Urban Food-Insecure
BBS & World Bank Slum Report

0% Digital
Current Traceability
Zero Audit Trail or OTP Safety



📸 The Reality: Millions of children in Dhaka lack access to a single hot meal daily

🌍 Real-World Impact Pillars (Why This System Matters):

🍛 **Dignified Hunger Relief (SDG 2):**
Transforms cafeteria surplus into 80–120 fresh hot meals daily for orphanages and street children with zero humiliation.

🛡️ **Food Safety & Legal Immunity:**
Strict BFSA thermal tracking protects donors and recipients; bad food is automatically rejected before pickup.

🌱 **Climate Action & Methane Prevention (SDG 12):**
Prevents tons of rotting organic waste in Dhaka city dumpsters, slashing lethal urban methane emissions.

Data Sources: UNEP Food Waste Index 2024, Bangladesh Food Safety Authority (BFSA), BBS Urban Slum Survey

2. Solution: The Core Idea & System Model

A synchronized 3-sided micro-logistics platform engineered for 15–30 minute rescue velocity

ONE SINGLE LINE (EXPLAINING THE PROJECT TO A KID OR PARENT)

CORE FORMULA

"An Uber-like app where university cafeterias post extra hot food, and nearby charities and student volunteers get alerted to pick it up and feed hungry children in 30 minutes before it gets cold!"

The 3-Sided Hyper-Local Ecosystem:



DONORS

1. Food Donors

Cafeterias, Caterers & Restaurants

- **60-Second Listing:** Title, quantity & cook time.
- **AI Spoilage Clock:** Automatic countdown dial.
- **Meal Converter:** '40 kg Biryani = 80-100 kids'.



RECIPIENTS

2. Verified NGOs

Orphanages & Local Shelters

- **Proximity Radar Map:** Filtered within 3-5 km.
- **1-Click Claim:** Instantly reserves surplus batch.
- **Digital QR Receipts:** Automated distribution audit.



COURIERS

3. Student Volunteers

Campus Cyclists & Bikers

- **Uber-Style Alerts:** Notified during commutes.
- **Fast Delivery:** 15–30 min pick & drop.
- **Insulated Bags:** Maintain thermal safety.

Ecosystem Synchronicity: Donor posts → NGO reserves → Student biker delivers → Handshake verified.

2. Main Features: What Will the End System Perform?

Four core algorithmic and operational modules solving real friction in food rescue



📸 Student Volunteer Squad • Rapid 15–30 Min Campus-to-Shelter Rescue

⏱️ 1. 60-Sec Post & AI Spoilage Countdown Meter

Donor inputs food title and cooking timestamp. The system calculates a dynamic biological expiration countdown based on BFSA parameters and flashes urgency rings.

📍 2. Proximity Radar Map (3 km / 5 km Radial Filter)

Bypasses notorious Dhaka gridlock by strictly clustering donations within walkable/cycling distance. Food under 2 hours flashes as '🚨 **Critical Priority**'.

🚴 3. Uber-Style Crowdsourced Student Volunteer Dispatch

Broadcasts micro-rescue notifications to university students commuting home on bicycles/motorcycles with an instant route view and thermal packaging checklist.

🔒 4. 4-Digit OTP & QR Code Anti-Fraud Protocol

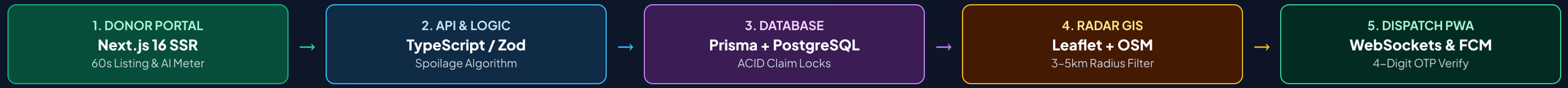
Zero leakage: The donor only releases food when the volunteer provides a secret 4-digit PIN. Shelter confirms final receipt via dynamic QR code scan.

System Output: Guaranteed hot, safe food delivered in under 30 minutes with 100% digital audit trace.

3. Plan of Action: High-Level Technology Overview

Modern open-source stack chosen for rapid response time, relational integrity, and zero hosting expense

SYSTEM ARCHITECTURE & DATA FLOW PIPELINE:



Frontend & PWA: Next.js 16

Why Chosen: Server-Side Rendering (SSR) delivers sub-100ms cold start times on 4G cellular data. Progressive Web App (PWA) enables instant volunteer dispatch without app store friction.

Data Layer: Prisma + PostgreSQL

Why Chosen: Strict ACID transactional guarantees prevent double-claiming of food batches. Type-safe Prisma models ensure zero runtime schema mismatch across the entire stack.

Spatial GIS: Leaflet.js + OSM

Why Chosen: 100% open-source mapping engine without expensive Google Maps API credit card billing. Performs fast client-side Haversine circular radius queries on low-spec phones.

Rationale: Zero cloud subscription costs, end-to-end TypeScript safety, and instantaneous mobile performance.

4. Project Timeline: 16-Week Agile Milestone Roadmap

Structured semester progression from system architecture to live UIU campus field pilot and defense

WEEKS 1 – 4

Phase 1: Architecture

- Comprehensive SRS documentation
- Prisma relational schema modeling
- 3-Portal responsive UI setup
- Role-based authentication

Status: 100% Completed

WEEKS 5 – 8

Phase 2: Core Engine

- AI Spoilage dynamic countdown
- Leaflet geospatial radar GIS (3km)
- 4-Digit OTP handshake logic
- Atomic claim transaction engine

Status: 90% Completed

WEEKS 9 – 12

Phase 3: Campus Pilot

- UIU Cafeteria live trial run
- Student cyclist volunteer onboarding
- Satarkul orphanage delivery test
- Turnaround latency optimization

Target: Live Field Run

WEEKS 13 – 16

Phase 4: Audit & Defense

- Concurrency & stress testing
- Anti-fraud security penetration test
- Project impact report & analytics
- Final SE Lab defense presentation

Target: Final Defense

🚀 **Ready to Transform Campus Food Surplus into Immediate Hunger Relief**

Zero commercial barrier • Real community impact • Rigorous Software Engineering execution

Q & A SESSION

Thank you! Questions are welcome.